TTC ScheduleWatch

Stop-Level Schedule Adherence Analysis of the TTC Streetcar Network Using Open AVL and GTFS Data

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https://github.com/neil-simmons/TTC-ScheduleWatch/tree/main

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Abstract

TTC ScheduleWatch is an open-data analysis and visualization tool that reconstructs the stop-level schedule adherence of the Toronto Transit Commission's streetcar network using publicly available Automatic Vehicle Location (AVL) and General Transit Feed Specification (GTFS) data. The TTC publishes system-level on-time performance figures measured only at route terminals, aggregating away the spatial and temporal structure that would make delay information actionable for planners, advocates, and riders. This project addresses that gap directly.

By projecting GPS vehicle traces onto GTFS route geometries and interpolating stop arrival times against a relative schedule baseline, the tool quantifies how delay originates and accumulates at the individual stop level across over 100,000 trips. A schedule signature framework groups trips sharing identical stop sequences and relative timings, enabling statistically valid comparisons across a large sample while maintaining mathematical consistency. Applied to seven weeks of TTC streetcar operations from March to May 2026, the tool identifies systematic deviation patterns, including abrupt reliability reversals within single corridors, time-of-day and day-of-week variation, and evidence of schedule baselines misaligned with actual operations. Equity indicators from the Statistics Canada 2021 Census are overlaid spatially to examine whether deviation severity correlates with neighbourhood-level income, transit dependence and other metrics. A schedule recalibration feature translates empirical arrival distributions into GTFS-compatible adjustment suggestions at the target percentile of the user's choice. The tool is deployed as an accessible web application requiring no technical background to operate, extending this analytical capability to researchers, transit advocates, and the public. The overarching concept for this project was conceived and implemented by the author. While AI tools were used in coding the functionality of project, all of the guiding logic, assumptions, interpretations of outcomes were made by the author.

1. Problem Statement

Delays in urban surface transit are not random. These unwanted deviations from schedule accumulate in specific places, at specific times, following patterns that may be predictable enough to plan around if only these data were available to impacted users - riders, advocates, professionals, and researchers. Currently, these data are not available in easy-to-use and interpret formats (Munro, 2025; Pizey-Allen, 2025).

The Toronto Transit Commission (TTC) publishes system-level on-time performance figures, but by aggregating metrics and measuring on-time performance at only terminals, the TTC averages away exactly the spatial and temporal structure that could make the information insightful (Furth et al., 2006; Toronto Transit Commission, 2025). A system or route-level on-time performance figure of 60% tells a planner very little about whether the delay is concentrated at three bottleneck stops or distributed evenly across the corridor, whether it is worse during the PM peak or consistent throughout the day. There is a gap in publicly accessible visuals and statistics to describe exactly where, when and how TTC streetcars on-time performance and schedule adherence evolves at the stop-by-stop level (Munro, 2025; Pizey-Allen, 2025; Wessel et al., 2017).

The published baseline schedule of trips is dependent upon by third-party applications as the basis for trip planning and arrival predictions when live information is not present (Google for Developers, 2022; Transit, n.d.). When riders use these platforms to plan journeys ahead of time, the underlying schedule is the reference. Delay patterns that are structurally embedded in TTC operations are therefore systematically misrepresented not just through the TTC's own channels, but across the digital tools riders use to interact with the network.

Understanding where schedule deviation accumulates and is largest is also a spatial equity question. TTC streetcar corridors pass through some of Toronto's most transit-dependent neighbourhoods.

To improve the way in which delays are quantified and displayed, this work seeks to identify, process, and analyze publicly available data. General Transit Feed Specification (GTFS) schedule data and Automatic Vehicle Location (AVL) records are both publicly available through the City of Toronto Open Data Portal. Together, they contain the information needed to visually track every TTC streetcar trip and calculate its detailed structure against a baseline schedule. Moreover, the tool developed for this project allows delay patterns to be examined alongside neighbourhood-level demographic data from the Statistics Canada 2021 Census (City of Toronto, 2026c), making it possible to ask whether the stops and corridors where schedule deviation is most severe are spatially correlated with the communities least able to absorb unreliable service.

TTC ScheduleWatch is an open-data analysis and visualization tool that reconstructs the spatial and temporal structure of TTC streetcar delay using only publicly available AVL and GTFS data. It quantifies schedule adherence at the stop level, visualizes how delay accumulates along corridors across individual trips, identifies the time periods and locations where deviation is most severe, and contextualizes those findings against neighbourhood equity indicators. A schedule recalibration feature translates the empirical schedule adherence distributions into objective adjustments to the published baseline schedule: one actionable output of the diagnostic capability the tool provides. This report describes the methodology underlying the tool, the data sources used, the findings produced when applied to seven weeks of TTC streetcar operations, and the limitations of the approach.

2. Methods

2.1 Data Integration

TTC ScheduleWatch is built on two publicly available datasets: archived AVL GPS vehicle location records from the City of Toronto Open Data Portal, and GTFS schedule and network data published by

the TTC. AVL data, comprised of latitude, longitude, and system time linked to specific route, vehicle and trip ID, were selected over alternative performance sources such as automated fare collection records, passenger surveys, or manual observation because it provides continuous spatial coverage across every trip and is available at no cost under an open government licence. GTFS establishes a machine-readable schedule baseline and structure to support trip identification, description, and plotting.

All spatial calculations are performed in UTM Zone 17N (EPSG:32617) rather than the geographic coordinate system WGS84 in which the source data arrives. UTM is a metric projection in which distance calculations are meaningful in metres, supporting computationally simpler analysis. Additionally, the GTFS data includes a shape_dist_traveled attribute, which indicates how far along a route a specific stop is located. However, the standard does not specify a unit of measurement. To ensure consistent units, the program detects and converts the values to kilometers. To achieve this, the program evaluates the maximum distance value of the route. Since no TTC streetcar route is greater than 500 kilometers long, a maximum value exceeding 500 cannot represent kilometers and is thus divided by 1,000, converting any meter values to kilometers.

2.2 Trip Filtering and Operational Day Definition

Raw AVL data requires several filtering steps before analysis can proceed. The transit operational day is defined as running from 4:00 AM to 3:59 AM the following calendar day. Trips that begin between 12:00 AM to 3:59 AM are assigned the previous date as their op_date. Known statutory holidays, specifically for this analysis period Good Friday, April 3, 2026, are treated as a distinct category.

GPS pings are filtered to retain only those falling within 150 metres of the GTFS route shape. This guards against positionally inaccurate GPS jitter or vehicles off route, while preserving valid GPS pings. Trip-day groups with fewer than three GPS observations are discarded entirely, as fewer than three pings cannot support meaningful interpolation across a multi-stop route.

2.3 The Schedule Signature

In order to enforce an apples-to-apples comparison across trips, the program defines a schedule signature. The schedule signature is a grouping of GTFS trips that share both an identical stop sequence and identical relative scheduled timing across every stop. This grouping is necessary because trips sharing the same route and direction can differ in their stop patterns or scheduled run times and thus comparing them against a single unified schedule reference would compare trips with differing operational expectations. By restricting each analysis to trips within a single signature, every observed run is measured against a mathematically identical schedule baseline that contains one relative schedule timing.

2.4 Temporal Framework

Time is converted to be measured relatively; it is the number of seconds elapsed from the trip's scheduled departure at its origin stop, rather than absolute clock time. This allows trips departing at different times of day, such as a 7:03 AM service and a 7:18 AM service operating under the same signature to be superimposed on the same time axis and compared against the same relative schedule baseline. Deviation from the baseline at any stop is then defined simply as the difference between the observed relative arrival time and the scheduled relative arrival time for that stop.

2.5 GPS Track Processing

Before stop arrival times can be estimated, each vehicle's GPS trace must be converted from a sequence of two-dimensional coordinates into a one-dimensional along-track distance profile. Each ping is projected onto the GTFS route shape and its distance along that shape recorded in kilometres. This representation is used as the spatial measurement throughout the tool, and it makes the slopes of time-distance traces directly interpretable as speed (Wessel et al., 2017).

Following projection, a rolling median filter with a window of three pings is applied to the resulting distance sequence. When a vehicle is stationary at a red light, GPS noise causes its projected distance to jitter slightly forward and backward. Without smoothing, a stationary vehicle may incrementally advance its position after applying monotonic enforcement. The rolling median stabilizes this scatter without discarding temporal observations.

Monotonically increasing distance is then enforced using cumulative maximum, constraining distances to never decrease. This removes GPS backtracking artifacts, potentially caused by signal error, without discarding the surrounding valid observations.

Terminal layover time is trimmed by identifying the last ping at the route's origin terminal distance and the first ping at the destination terminal distance, then slicing the trace strictly between those indices. Vehicles frequently sit at terminals for several minutes before departing or after arriving and including that stationary time would stretch the generated charts unreasonably. Further, when a user selects a sub-corridor rather than the full route, a 0.1-kilometre buffer is added beyond the first and last stops of the selected range to ensure that pings immediately surrounding those boundary stops are not excluded from interpolation.

2.6 Stop Arrival Estimation

Stop arrival times are estimated by identifying, for each GTFS stop, the pair of GPS pings with along-track distances bracketing the stop's known position on the route shape. Interpolation uses a left-biased array search, which targets the moment the vehicle first reaches the stop's physical boundary rather than the mid-point of any dwell period or the departure moment. This ensures all measurements are strictly arrival metrics.

Linear interpolation between the bounding pings yields an estimated arrival time in relative seconds (Furth et al., 2006; Wessel et al., 2017). If the two bounding pings are separated by more than 60 seconds, the estimate is nullified. A gap of that duration means the vehicle could have travelled several hundred metres without observation, making any interpolated position unreliable. This is a data quality gate rather than an analytical assumption.

When the "Align to First Observed Stop" mode is active, the trip's relative timeline is anchored to the second stop in the sequence rather than the start terminal. While technically the first stop is the scheduled origin, vehicles frequently begin GPS tracking while already in motion, meaning the first stop may lack a valid GPS observation. The second stop is the first position where a reliable stationary ping can be observed and interpolated. This feature is intended to clean up the data to isolate deviation across the route, rather than at the terminal. If the GPS gap surrounding this anchor stop exceeds 60 seconds, the trip run is rejected entirely as anchoring the full relative timeline of a run on a very uncertain interpolation would propagate that uncertainty across every subsequent stop. This mode is automatically disabled when the user selects a sub-corridor whose start stop is not the true route origin, since anchoring to a mid-route stop would erase the delay accumulated over the preceding stops, invalidating the relative schedule reference.

When consecutive pings within a trip trace are separated by more than 60 seconds, a null break is injected into the rendered line. This produces a visible gap in the time-distance chart rather than a misleading continuous line drawn across a data void.

2.7 On-Time Performance Measurement

Schedule adherence is measured at the individual stop rather than aggregated to the trip or route level. Stop-level measurement reveals where along a corridor schedule deviation originates and accumulates.

On-time performance at each stop is defined as the percentage of observed arrivals falling within a configurable early-late window relative to the scheduled arrival time. The default window of −60 to +120 seconds reflects a chosen tight rider tolerance for early and late arrivals. However, since this default is tighter than industry standard, the window is exposed as a user-configurable range of -300 to +300 seconds (-5 minutes to + 5 minutes) to facilitate different operational contexts and analytical purposes requiring different thresholds (Board et al., 2013; Furth et al., 2006).

In addition to displaying stop-by-stop reliability, the tool calculates whether on-time performance drops more than ten percentage points between two consecutive stops, and if it does it flags this as ‘reliability cliff’ to the user. Gradual deviation along a corridor is expected of surface transit operations; the ten-point threshold isolates large changes that may be attributable to a more specific physical or temporal event.

Stops with fewer than three observations are excluded from the reliability map and recalibration output. The minimum for the best-performing stop insight is raised to five to reduce the influence of near-empty samples on a prominently displayed metric. The density chart applies a tiered colour-coding, red below ten observations, amber below twenty-five, and blue at twenty-five and above, to visually communicate the confidence level attached to each stop's estimated distribution. This is in addition to the prominently displayed number of trips processed in the dashboard/insights panel.

2.8 Analytical Decomposition

Six analytical views support interpretation of the stop-level reliability findings. The day-of-week chart groups the mean delay of individual trips into box-and-whisker plots for each day of the week, highlighting differences in operational conditions between weekdays and weekends. The daily trend chart tracks the daily median deviation across the entire analysis period. To smooth out minor day-to-day fluctuations and reveal weekly performance cycles, the chart applies a seven-day rolling mean, which is computed only when the selected query spans at least seven distinct operating dates. The departure-hour scatter plot fits a first-order linear regression to the relationship between trip departure hour and median trip delay, summarizing whether deviation worsens or improves across the selected period. A higher-order polynomial fit was considered but rejected as the feature is intended to show basic context.

The delay distribution by stop box plot displays the complete distribution of arrival deviations at each sequential stop along the route, making it possible to identify where travel times begin to exhibit high variance and unpredictability. The stop-by-time-of-day heatmap displays median stop deviations across six specific service periods: Early AM (4:00 to 7:00), AM Peak (7:00 to 10:00), Midday (10:00 to 14:00), PM Peak (14:00 to 18:00), Evening (18:00 to 21:00), and Late Night (21:00 to 3:59). To maintain statistical reliability, cells representing stop-period combinations with fewer than three observations are left blank. The stop-by-day-of-week heatmap cross-references each physical stop against the days of the week, displaying the median arrival deviation to reveal whether specific localized delays are day-dependent or persistent throughout the entire weekly schedule.

Two time window filtering modes allow users to define which trips are included in an analysis. Trip Start Mode includes only trips that originate within the selected time window, isolating specific scheduled blocks such as AM Peak departures. Overlap Mode includes any trip whose trace touches the selected window regardless of when it departed, which is appropriate for measuring corridor conditions at a specific time of day across all vehicles present during that period.

2.9 Schedule Recalibration

Where a stop has at least three observations, the tool computes a suggested adjusted arrival time by using a basic approach of taking the Nth empirical percentile of the observed relative arrival time distribution and shifting the current GTFS scheduled time by the corresponding offset (Cevallos et al., 2011). The percentile-based approach was chosen over adjustment by solely median delay because it directly controls the fraction of future trips that will not arrive over 0 second late under the revised schedule. Setting the

target to the 50th percentile means approximately 50% of trips will not arrive late over 0 seconds. The 50th percentile is set as the default. The slider is exposed from the 50th to the 95th percentile. Output is formatted as a GTFS-compatible stop_times reference file for a single analyzed signature.

2.10 Route-Level Aggregation

When the option to run an analysis across multiple signatures at once by headsign/route is chosen, the program uses the exact same mathematical model as normal, computing in a loop every unique signature's reliability in every selected route/headsign. This mode, however, filters any headsigns including "short" as short-turn operations could not be confirmed as representing pre-planned scheduled modifications or reactive operational decisions. Yet, this process is still computationally expensive and is locked to one user at one time across the server. This mode also disables most other analytical charts that are based on single-signature analysis.

Further, stops served by more than one route are aggregated using a sample-size-weighted micro-average. The total number of on-time observations across all routes is divided by the total number of observations. This prevents a route with two observed runs from exerting the same mathematical influence as a route with fifty at a shared stop, reflecting a more accurate rider experience. In the map view, routes sharing a physical corridor are rendered with a perpendicular spatial offset of 5 metres plus 8 metres per additional route index, applied in UTM space, to prevent overlapping geometries from overlapping each other at standard map zoom levels.

2.11 Equity Analysis

Neighbourhood-level demographic indicators from the Statistics Canada 2021 Census, accessed via the City of Toronto Neighbourhood Profiles dataset (City of Toronto, 2026c), are overlaid spatially with stop-level reliability results. Individual-level data is not used. Six indicators were selected: median household income, low-income household percentage, transit commute share, visible minority population share, recent immigrant share, and seniors 65 and older (Allen & Farber, 2020). These attempt to capture economic vulnerability, transit dependence, and varying demographic groups. Each stop is assigned to a neighbourhood via a point-in-polygon spatial join. The resulting scatter plots and map layers allow patterns in deviation severity to be examined against neighbourhood context.

3. Data Sources

The data used in this project comes solely from the City of Toronto Open Data Portal. The following datasets are used. Merged GTFS - TTC Routes and Schedules from March 15 to May 2, 2026, is used to integrate all the schedule data with AVL data (City of Toronto, 2026b). The AVL data was harvested using ttc_harvestor.py in the GitHub repository almost until May 2. The AVL dataset used is TTC BusTime Real-Time Next Vehicle Arrival (NVAS) (City of Toronto, 2024).

To contextualize the GTFS and AVL data, the Neighbourhoods dataset was combined with the Neighbourhoods Profile dataset using precompute_equity.py to precompute the base equity context to be layered with the custom analysis by the user (City of Toronto, 2026a, 2026c).

To have a usable network on load for users, generate_precomputed_map.py was used to process the data for a specific analysis containing all-day on weekdays on all routes. Custom analyses do not depend on this process.

All the data is hosted on HuggingFace at https://huggingface.co/datasets/neil-simmons/ttc-avl-data. The programs are hosted on GitHub. More information about the datasets contained on HuggingFace is provided in the Data Sources Statement.

The following chart summarizes the attributes used within the datasets.

Table 1: Dataset Attributes

Dataset	File	Attributes
Merged GTFS - TTC Routes and Schedules	trips.txt	route_id
		trip_id
		trip_id
		trip_headsign
	stops.txt	stop_id
		stop_name
		stop_lat
		stop_lon
	stop_times.txt	trip_id
		stop_id
		arrival_time
		stop_sequence
		shape_dist_traveled
	shapes.txt	shape_id
		shape_pt_lat
		shape_pt_lon
		shape_pt_sequence
TTC BusTime Real-Time Next Vehicle Arrival (NVAS)	ttc_all_streetcars_history.parquet	route_id
		trip_id
		system_time
		latitude
		longitude
Neighbourhoods and Neighbourhoods Profiles	equity_neighbourhoods.geojson	area_name
		median_income
		low_income_pct
		transit_commute_pct
		visible_minority_pct
		recent_immigrant_pct
		senior_pct
		geometry

4. Results

Applied to over 100,000 TTC streetcar trips across seven routes over 35 weekday operating dates between March 15 and May 2, 2026, TTC ScheduleWatch processed thousands of stop-level arrival estimates across 642 unique stops from stops.txt. System-wide on-time performance under the default −60 to +120 second window averaged 30.2% across all analyzed stops on weekdays throughout the entire operating day, with individual stop performance ranging from 0% at the worst-served locations to 91% at the best. Custom analyses are highly configurable, leading to a wide variety of results. The figures below are selected to illustrate the tool's analytical capabilities across several routes and configurations.

4.1 Schedule Misalignment on Route 512

Route 512 was chosen as it was the starkest example found of schedule misalignment across the network. The 7:00 – 9:00 Weekdays time interval was chosen as it is representative of a typical rush hour.

Figure 1a

Route 512 East, Weekdays, 7:00-9:00, Reliability Windows -60s to +120s Density Chart, 506 Trips from the Schedule Signature Ranging 5:30AM to 9:30AM.

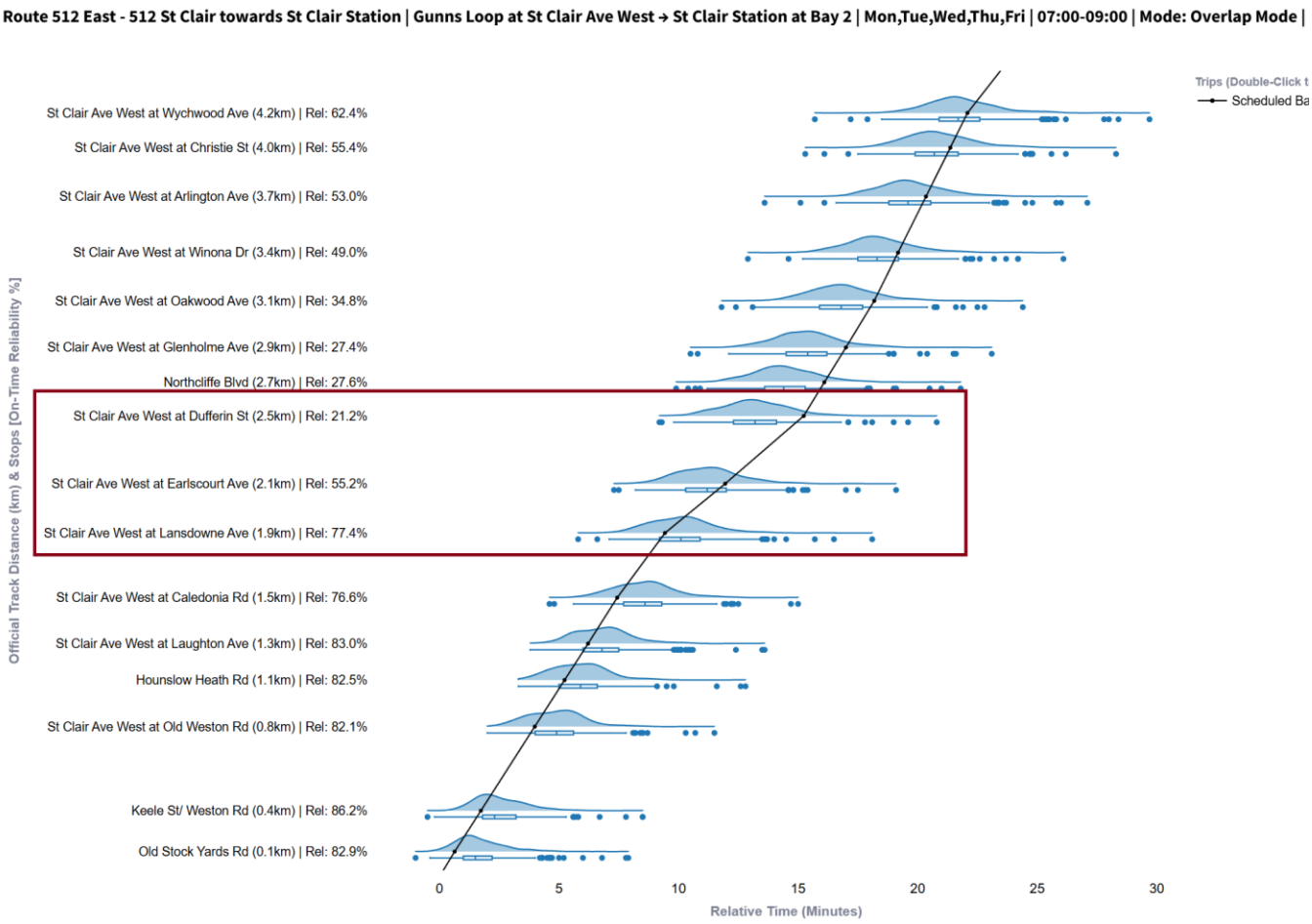


Figure 1b

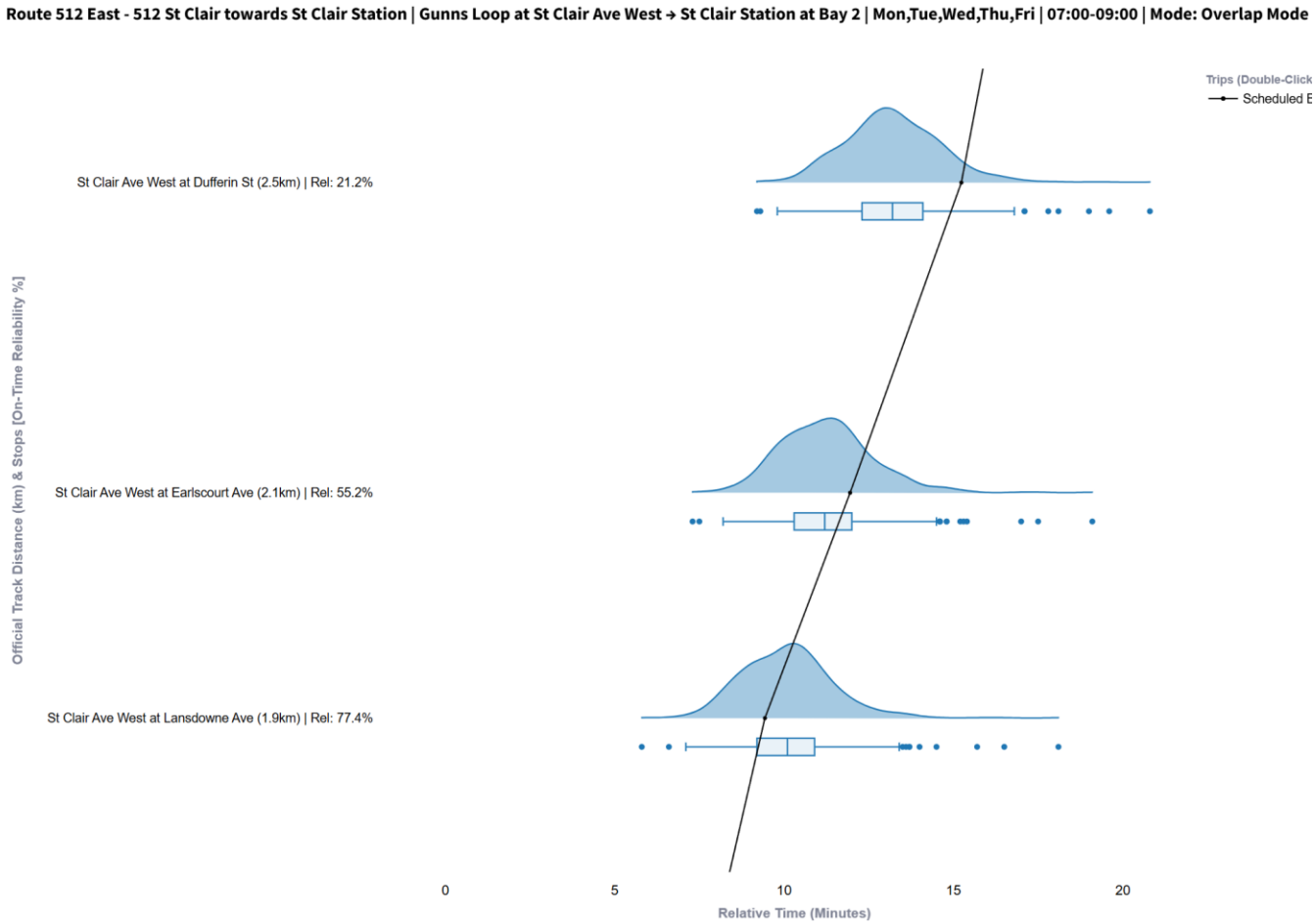


Figure 1, without terminal alignment, shows a pronounced reliability reversal within two consecutive stops: from 77.4% on-time at St Clair Ave West at Lansdowne Ave to 21.2% at St Clair Ave West at Dufferin St. This is a reliability cliff by the tool's definition, a drop exceeding ten percentage points between adjacent stops, and locates a specific bottleneck in the AM peak corridor with some precision.

Figure 2

Route 512 East, Weekdays, 7:00-9:00, Reliability Windows -60s to +120s, t=0 Aligned, Time-Distance Chart, 490 Trips from the Schedule Signature Ranging 5:30AM to 9:30AM.



Figure 2 applies the second-stop alignment, anchoring each trip's relative timeline to the first reliably observed stop rather than the scheduled terminal departure. Under this configuration, the same corridor's reliability drops from 72.9% to 3.5% at the same transition point. The near-total collapse under alignment is the more informative result. It indicates that the current GTFS schedule baseline is structurally offset from how vehicles actually depart terminals, and that measured reliability in the unaligned view is partly an artifact of trips being systematically early from the origin. Once that terminal offset is removed, almost no trips are meeting the schedule within the defined window at the worst-performing stop.

4.2 Time-of-Day and Day-of-Week Variation on Route 506

Figure 3

Route 506 West, Weekdays, 00:00-23:59, Reliability Windows -60s to +120s, Median Delay – Stop x Time of Day, 651 Trips from the Schedule Signature Ranging 9:39AM to 6:24PM

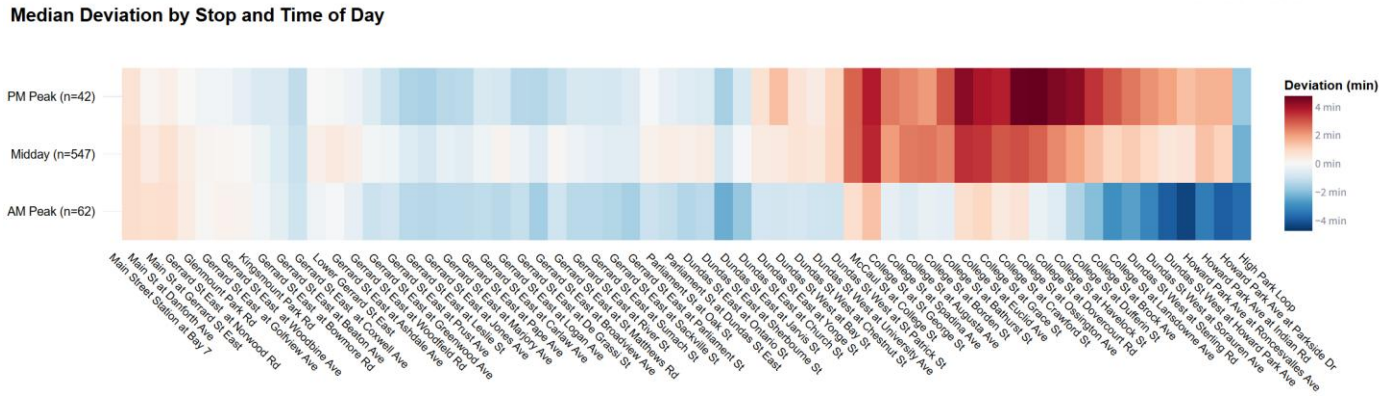


Figure 3 examines Route 506 westbound across the full operating day, plotting median arrival deviation by stop and service period. AM peak trips (n=42) tend to arrive early relative to schedule, PM peak trips (n=62) tend to arrive late, and midday trips (n=547) span the largest sample and show intermediate performance. The asymmetry between morning and afternoon patterns, combined with the heavily unequal trip counts per period, suggests the current schedule may be calibrated around a midday operational baseline that does not reflect peak conditions in either direction. The schedule might have potential to be disaggregated by time period.

Figure 4

Route 506 West, Weekdays, 00:00-23:59, Reliability Windows -60s to +120s, Median Deviation by Stop and Day of Week, 651 Trips from the Schedule Signature Ranging 9:39AM to 6:24PM

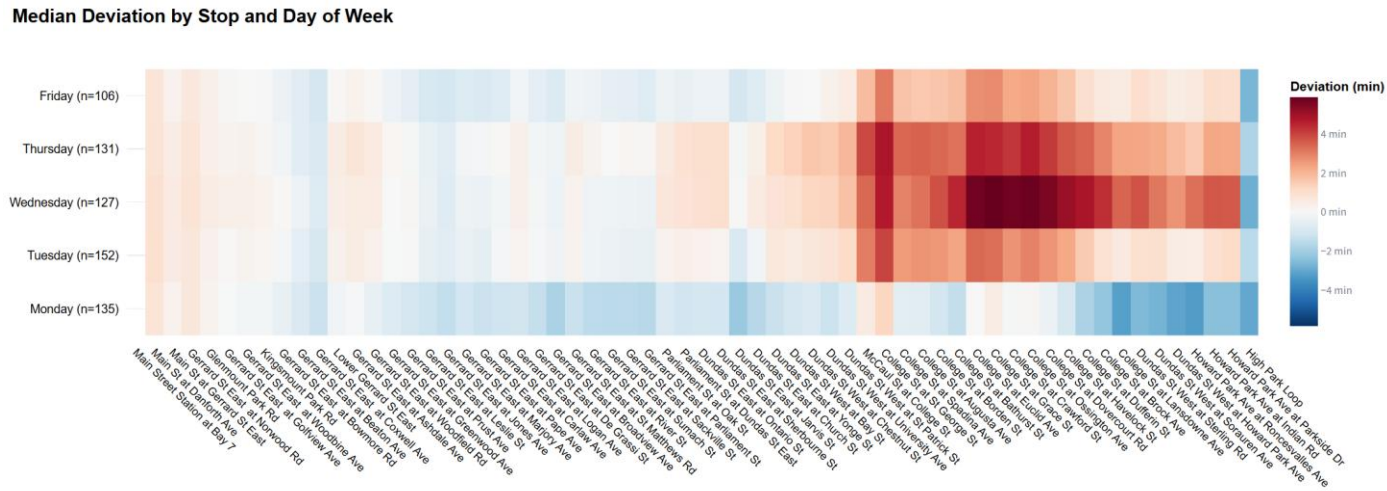


Figure 4 presents the same route broken down by day of week. Monday and Wednesday show notably different contrasting deviation profiles at the same stops relative to Tuesday, Thursday, and Friday. This day-level variation, consistent enough to be visible in the heatmap across a multi-week sample, is further evidence that a single weekly schedule is averaging over distinct operating conditions.

4.3 Temporal Trend on Route 510

The following charts were selected simply to highlight interesting findings from the dataset.

Figure 5

Route 510 North, All Days, 00:00-23:59, Reliability Windows -60s to +120s, Daily Median Deviation Over Analysis Period, 370 Trips from the Schedule Signature Ranging 9:39AM to 2:29PM

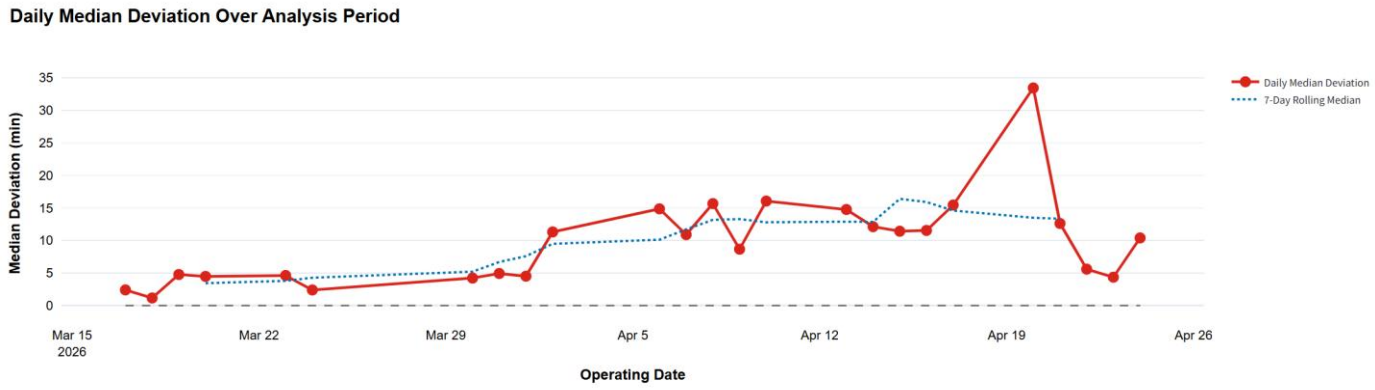


Figure 5 tracks the daily median deviation for Route 510 northbound across 370 trips. Deviation increases through April relative to March, with a notable spike on April 20th that stands apart from the seven-day rolling trend. The tool only surface these patterns for investigation, but does not attempt to draw conclusions on origins.

4.4 Departure Hour Pattern on Route 11

Figure 6

Route 511 North, Weekdays, 00:00-23:59, Reliability Windows -60s to +120s, Per-Trip Mean Delay vs Departure Hour Scatter Plot, 370 Trips from the Schedule Signature Ranging 6:44AM to 11:24AM

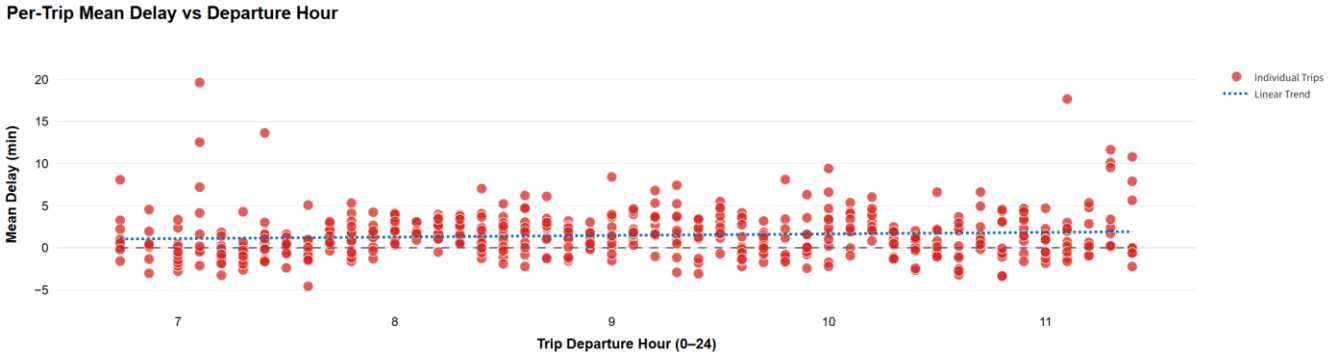


Figure 6 plots per-trip mean delay against departure hour for Route 511 northbound across all weekdays. Rather than a monotonic trend, the scatter exhibits a wave-like pattern across the hour cycle. This could reflect bunching dynamics or systematic interactions with non-TTC traffic at specific points in the hour.

4.5 Schedule Recalibration on Route 512

This specific route 512 configuration was chosen as it demonstrates significant schedule deviation that, when aligning to median, results in large minute shifts in the schedule.

Figure 7

Route 512 West, Weekdays, 00:00-23:59, Reliability Windows -60s to +120s, Schedule Recalibration, 1049 Trips from the Schedule Signature Ranging 2:28PM to 7:24PM

Stop	Current GTFS Time	Suggested Time	Adjustment (min)	Observations
St Clair Ave West at Glenholme Ave West Side	14:51:27	14:53:05	1.6	1027
St Clair Ave West at Northcliffe Blvd West Side	14:52:18	14:53:55	1.6	1027
St Clair Ave West at Dufferin St West Side	14:53:08	14:55:11	2.0	1024
St Clair Ave West at Earlscourt Ave	14:54:05	14:56:35	2.5	1026
St Clair Ave West at Lansdowne Ave West Side	14:55:11	14:58:22	3.2	1010
St Clair Ave West at Caledonia Rd West Side	14:56:41	14:59:40	3.0	949
St Clair Ave West at Laughton Ave West Side	14:57:43	15:00:48	3.1	942
St Clair Ave West at Silverthorn Ave West Side	14:58:34	15:01:48	3.2	936
St Clair Ave West at Old Weston Rd West Side	14:59:39	15:03:08	3.5	935
St Clair Ave West at Weston Rd/ Keele St West Side	15:01:18	15:04:54	3.6	932

Figure 7 presents the schedule recalibration output for Route 512 west on weekdays, adjusted to the 50th percentile of observed arrival times. The displayed adjustments range from 1.6 to 3.6 minutes added. The recalibrated schedule times would be centered around the measured distribution of arrival times, a meaningful improvement for third-party applications that plan ahead using GTFS.

4.6 Route-Level Aggregation and Equity Context

Figure 8

All-Day Weekdays, All Routes, Lower-Income Households Associated with Lower On-Time Reliability Map View

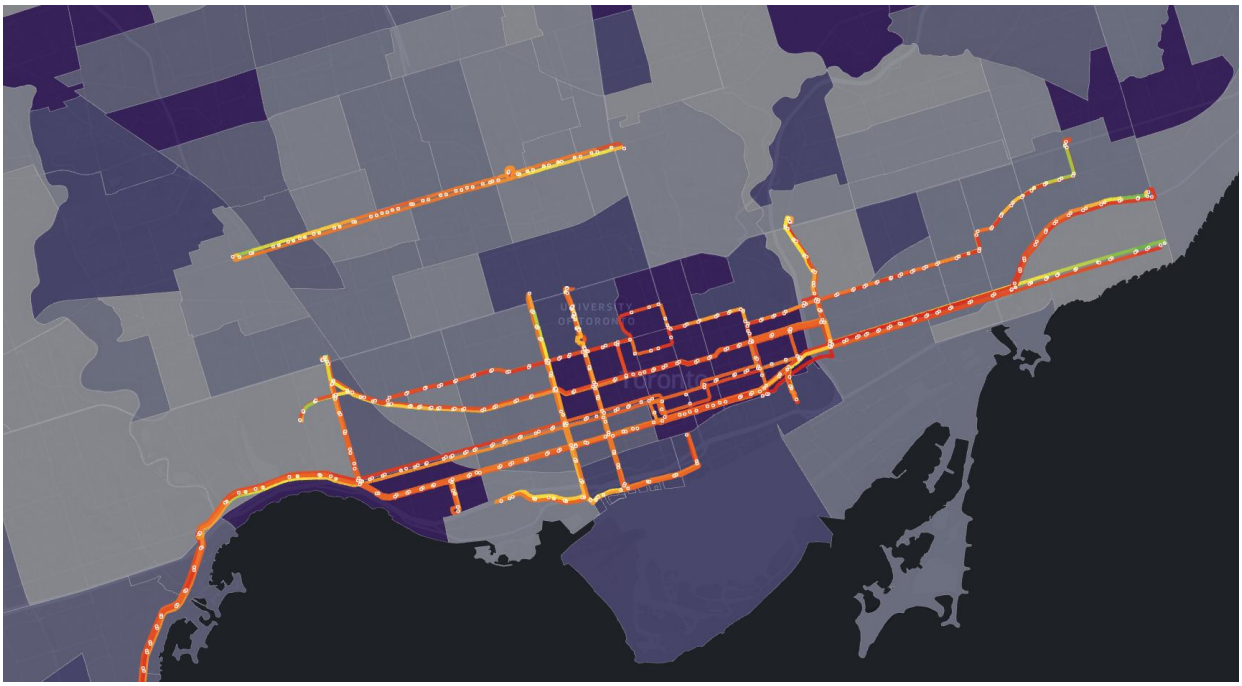
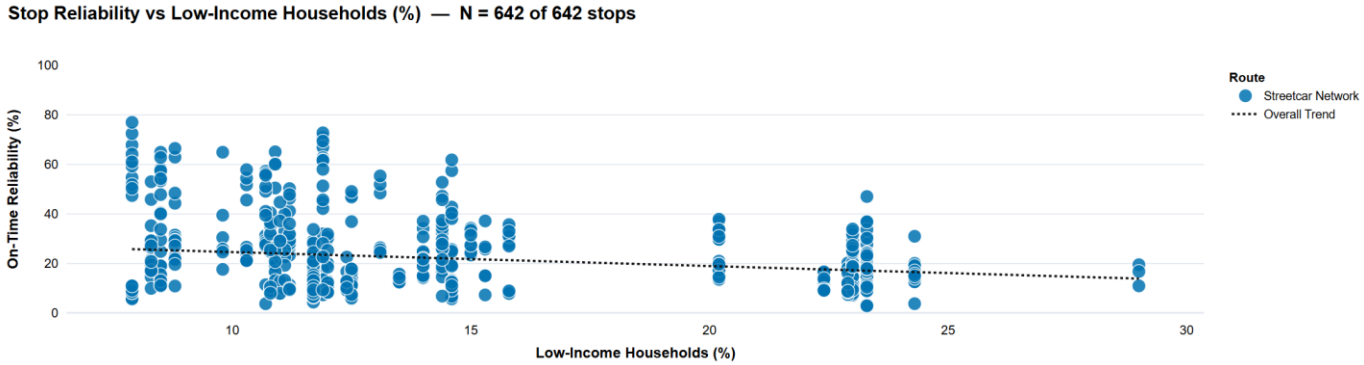


Figure 9

All-Day Weekdays, All Routes, Lower-Income Households Associated with Lower On-Time Reliability Scatter Plot



Figures 8 and 9 present the all-day weekday analysis across all routes, with stop reliability mapped against the low-income household share of each stop's neighbourhood. The map view, Figure 8, shows the spatial distribution of on-time performance across the streetcar network overlaid on neighbourhood income shading, while the scatter plot, Figure 9, plots stop reliability directly against low-income household percentage across all 642 stops with sufficient observations. A negative trend is visible across the full sample, suggesting that stops in lower-income neighbourhoods tend toward lower on-time performance, though the relationship shows substantial spread.

5. Limitations

5.1 Measurement Precision and Data Quality

Estimating stop arrival times, the tool's core output, is derived from interpolation between GPS observations rather than direct measurement. AVL systems on the TTC streetcar network poll approximately every 15 to 30 seconds in the dataset, meaning a vehicle can travel a substantial distance between recorded positions. Linear interpolation between bounding pings assumes constant speed across the segment. This calculation distributes travel time evenly, so if a streetcar is held at a red light immediately before a stop and then accelerates through it, the interpolated arrival time may be earlier or later than the true event. The tool does not consider dwell times. The 60-second gap filter removes the most egregious cases, but interpolation uncertainty is present at every stop and stop-level timing should be understood as approximations rather than exact observations. This is the key reason for attempting to aggregate as many samples under one schedule signature as possible and why the trip sample size is displayed prominently on the dashboard.

Confidence in individual stop estimates varies substantially with sample size, but the tool does not propagate this uncertainty formally into its summary statistics with uncertainty bounds. The program includes only colour-coded sample size indicators in the density chart, observation counts per-stop in the time-distance chart and the observation counts in the dashboard.

Performance is also measured against the published GTFS schedule as a fixed ground truth. If the GTFS contains errors, has not been updated to reflect a recent service change, or includes scheduled times that were never operationally intended to be achieved, the tool will measure deviation against a flawed baseline.

5.2 Survivorship Bias

The analysis pipeline applies several filters that remove trip runs from consideration: pings outside the 150-metre corridor, trips with 60-second interpolation gaps at key positions, trips starting mid-route, and short-turn trips. Since runs that are filtered out are not explicitly counted or reported, this creates a form of survivorship bias where the reliability metrics produced by the tool describes only the trips that survived all quality gates, not the full operational picture. If highly deviated runs which so happen to be correlated with degraded GPS coverage or reactive short turns are disproportionately removed by these filters, the resulting reliability figures may be modestly optimistic relative to the true distribution of service quality.

5.3 Schedule Signature Fragmentation

The strict definition of a schedule signature, identical stop sequence and identical relative timing across all stops, can fragment what operationally constitutes a single service pattern into several distinct signatures when minor GTFS schedule variations exist across trips. A planner analyzing a route may need to run the analysis separately across multiple signatures and aggregate the findings manually. The tool mitigates this by providing descriptions, stop counts, and trip counts for each detected signature, allowing users to identify and prioritize the dominant pattern. The program also provides detailed analysis charts and tools up to the individual trip-level to allow for deep analysis of the nature of every signature. However, it does not resolve the fragmentation of signatures automatically.

5.4 Temporal and Baseline Scope

The dataset covers approximately seven weeks of TTC streetcar operations, from March 15 to May 2, 2026. This captures a single operating period in spring, excluding the seasonal variation such as winter conditions, summer construction, major events, school-year ridership differences, and the schedule changes that accompany each new GTFS release. Findings should not be generalized as representative of annual TTC performance; they describe what was happening during this specific window.

Within that window, statutory holidays are identified manually. Only Good Friday, April 3, 2026, is flagged as a holiday in the current implementation. Other special event days or unscheduled service modifications within the analysis period are processed as regular service, which may introduce noise into day-of-week and time-of-day comparisons for dates where operations deviated from standard patterns.

The tool does not currently support trips outside of the March 15 to May 2 range as that would require matching different groups of trips.txt, routes.txt, stop_times.txt, shapes.txt within the GTFS dataset to their active dates, however, this improvement is feasible as the GTFS dataset includes a feed_info.txt file that specifies the active dates of the schedules. In this way, year-round AVL data can be matched with GTFS schedules and analyzed.

5.5 Equity Analysis Scope

The equity analysis is descriptive and spatial. Stop-level reliability figures are compared against neighbourhood-level Census indicators from the Statistics Canada 2021 Census via a point-in-polygon spatial join, assigning each stop to whichever neighbourhood polygon contains it. Stops falling on neighbourhood boundaries may be assigned to either side depending on coordinate precision. The analysis does not produce or imply causal relationships. The neighbourhood-level aggregation also hides within-neighbourhood diversity; a corridor stop and a residential interior stop may fall in the same neighbourhood polygon while serving different communities.

Further, each stop considers only the equity factors of the neighbourhood its in, not the demographics of the passengers that use the stop who may have origins and destinations within or outside the neighbourhood.

5.6 Analytical Scope

The reliability and delay statistics are calculated on a per-vehicle basis. Because automated passenger counter or fare card data is not integrated, a delayed streetcar carrying 150 passengers during the AM Peak carries the same statistical weight as a nearly empty vehicle delayed during the late-night period. The tool measures operational schedule adherence; it does not measure the delay burden actually experienced by the riding public (Furth et al., 2006). A complete picture of passenger impact would require weighting each observation by the number of riders affected, data which is not available in the open sources used here.

More fundamentally, schedule adherence is one dimension of transit reliability, but arguably not the most important one from a rider perspective. For high-frequency transit like TTC streetcars, the interval between consecutive vehicles, headway regularity, is often central to a rider's experience and wait-time. A streetcar running several minutes behind its scheduled time is less disruptive to a rider if the following vehicle is also running late and maintaining a consistent gap. The tool does not directly measure headway regularity or vehicle spacing as the measured 15-30 second frequency of the AVL data introduces significant uncertainty that could not be averaged out in the same way of stop arrival times. Instead, the tool should be interpreted as revealing where, when and how schedules are not being met, identifying specific variables that may be linked to schedule deviation.

The schedule recalibration output carries a related caveat. The suggested stop_times.txt file covers a single analyzed signature and is explicitly labelled as a reference rather than an importable GTFS feed. Translating these suggestions into an operational schedule change would require a planner to reconcile the adjusted times against the full GTFS service structure.

6. Conclusions

TTC ScheduleWatch set out to address the specific gap that the spatial and temporal structure of TTC streetcar schedule deviation is not visible through publicly available data, and that existing performance figures average away precisely the stop-level detail that provides for actionable insights. The project demonstrates that this gap can be substantially closed using only open data. AVL records and GTFS schedules, both available through the City of Toronto Open Data Portal, contain sufficient information to reconstruct how delay originates and accumulates along corridors, identify where and when performance is worst, and produce empirically grounded schedule adjustment recommendations.

Several directions would strengthen and extend the tool. Higher-frequency AVL data would reduce interpolation uncertainty at closely spaced stops and support headway adherence analysis, completing the reliability picture from a rider perspective. Formal confidence intervals on the recalibration output would better communicate which suggested adjustments are well-supported by data and which require more observations. Support for multi-signature selection could increase sample sizes on deviation charts not tied strictly to one schedule baseline. The most significant opportunity, however, is extension to the TTC bus network. The methodology used in this tool is not inherently streetcar-specific, and the same pipeline applies to buses once the substantially larger data volume can be accommodated.

That the stop-level spatial and temporal structure of TTC schedule deviation, not present in the agency's own public reporting, can be reconstructed by a student using freely available open data reflects both what modern open data standards make possible and what transit agencies could choose to communicate, but have not yet chosen to.

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